Ziphozihle Mqopi

30/30

100,10

Wire ways and wiring,
FORMATIVE ASSESSMENT

Module #	671101000-KM-05:
NQF Level	level 4,
Notional hours	110
Credit(s)	Cr 11
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ **points**. In order to meet the pass mark, learners are required to achieve a minimum of **80%** of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓

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Barriers

People to be involved with assessment

Learner: Ziphozinle Maopi	Manager:
Trainer:	Mentor / Coach:
Facilitator: Balennah Tjabadi	
Assessor: Balennah Tjabadi	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Ziphozinle Maopi

Learner's signature: _____ Date:

Facilitator Name: Balennah Tjabadi

Facilitator's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
   - If the laid down assessment procedures were not followed during assessments
   - If not all evidence available was taken into account during the assessment
   - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
   - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
   - Not all the range items were available for assessment

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TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

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PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate **and put them at ease**	✓
2	**Explain the purpose of the meeting** (why you are there and how long the meeting will take)	✓
3	**Explain the** ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	**Explain** ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	**Explain the assessment process?** ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	**Give Learner copies of the following documentation and explain each document** ▪ The Assessment Guide which includes o The relevant unit standard (s) o Assessment contract o Assessment plan o Observation checklist o Knowledge checklist	✓
7	**Discuss the assessment plan** (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	**Tell the candidate his/her rights and responsibilities, the assessment procedures and policies** ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	**Ensure the assessment environment is appropriate** or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

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Complete the following questions as per instructions provided

EASTC
TECHNOCENTRIC
VARSITY

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support and fixing: Wireways must be securely fixed to prevent movement and ensure stability. They should be supported at intervals not exceeding 1.5 ~~metre~~ meters to maintain structural integrity

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b.)

i) Cable Type and size: The type of cables being used for instance power, control, data, or fiber optic and their dimestions significantly influence wireway selection. The wireway must be ~~reparbie~~ capable of accommodating the larger cable diameter and sufficient quantity of cables

2) Environmental Conditions: The installation environment impacts material selection. factors include temperature extremes, humidity, exposure to corrosive substances and pontential physical damage. For outdoor applications, weather resistant materials may be required.

3) Material: Wireways are made from various materials including steel, aluminium, and fiberglass. The material choice will depend on enviromental conditions, weight considerations and whether EMI (Electromagnetic Interference) shielding is needed.

4) Load Capacity: The wireway must support weight of cables within it. This includes the static load of cables as well as any dynamic loads from installation methods.

5) Installation Method: The method of installing the wireway for example wall-mounted, ceiling-mouted or free standing will influence the design and dimenstions required.

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Task	Questions Description	Marks
3	Describe the purpose of load distribution, lightning arrestors and energy control units.	5

3)

i) Load distribution

The purpose of load distribution is to distribute electrical power safely and efficiently to various circuits / loads, ensuring balanced loading. It prevents overloads, voltage drops and ensures reliable supply to equipment. Typically involves distribution boards, switchgear and circuit management.

2) Lightning Arrestor / Surge arrestor: It plays a role in protecting equipment from lightning-induced surges and voltage spikes. It diverts surge currents to ground, safeguarding sensitive electronics and equipment. It is installed at critical points like substations, buildings or equipment inlets.

3) Energy control units:

The purpose of energy control units is to monitor and manage energy usage, often providing control and protection. It enables efficient energy use, load shedding, remote control and sometimes billing. Energy control meters can include smart meters, energy meters or building managements systems. It is often integrate with automation system for enhanced functionality. It also helps manage peak demand, reduce costs and improve system reliability

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Task	Questions Description	Marks
4	Explain the various processes of measuring, testing and calculating earthing and bonding system values.	5

i) Earth Resistance Testing:

- Measures resistance of earth electrode to ground.

The methods include Fall-of-pontential Wenner Array, or clamp-on testing. It ensures earthing system can handle fault currents safely.

2) Earth Loop Impedance Testing:

- Measures impedance of earth fault loop (phases to earth). It ensures protective devices (fuses, circuit breakers) operate within safe times. It ~~typi~~ is typically done using loop impedance testers.

3) Continuity Testing

- Checks bonding and earthing conductors continuity. It uses low-resistance ohmmeters to verify connections. It ensures a safe path for fault currents

4) Soil Resistivity Testing

- Determines soil resistivity (Ωm) using Wenner or Schlumberger methods. It helps design effective earthing system by assessing ground conditions. Crucial for designing earthing grids or electrodes

5) Calculation and Analysis: calculates earth fault levels step voltages and earth pontential rise. Compare results with standards for instance SANS, IEC to ensure safety and compliance.

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ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS
Good work

ASSESSMENT JUDGEMENT		
Learner's Total Mark: 100	**Requirements met** ☒	**Requirements not met** ☐
Action/s required: **By when:**		

LEARNER FEEDBACK & COMMENTS
Good enough

DECLARATION BY THE FACILITATOR
I Baienah Tjabadi (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

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Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

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Assessment Planning

	Theory	Practical
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments

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